EXAM 2

BLAKE FARMAN

Lafayette College

Answer the questions in the spaces provided on the question sheets and turn them in at the end of the exam period.
It is advised, although not required, that you check your answers. All supporting work is required. Unsupported or otherwise mysterious answers will **not receive credit.**
You may **not** use a calculator or any other electronic device, including cell phones, smart watches, etc.

Name: Solutions

Functions

1. Find the value of b that makes the function

$$f(x) = \begin{cases} \dfrac{x^2-9}{x+3} & \text{if } -3 < x \\ 3x+b & \text{if } x \leq -3 \end{cases}$$

a continuous function.

$$\lim_{x \to -3^+} f(x) = \lim_{x \to -3^+} \frac{x^2-9}{x+3} = \lim_{x \to -3^+} \frac{(x+3)(x-3)}{(x+3)} = \lim_{x \to -3^+} x-3 = -3-3 = -6$$

$$\lim_{x \to -3^-} f(x) = 3(-3)+b = -6+b$$

$$-6 = -6+b \Rightarrow \boxed{0 = b}$$

Date: April 1, 2019.

Derivatives

2. Use the **limit definition**

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

to compute the derivative of

$$f(x) = x^2.$$

$$\lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \to 0} \frac{(x+h)^2 - x^2}{h}$$

$$= \lim_{h \to 0} \frac{x^2 + 2xh + h^2 - x^2}{h}$$

$$= \lim_{h \to 0} \frac{2xh + h^2}{h}$$

$$= \lim_{h \to 0} 2x + h$$

$$= \boxed{2x}$$

Derivative Rules

3. Compute the line tangent to

$$f(x) = \sin(x)$$

at the point $(\pi/2, 1)$.

$f'(x) = \cos(x)$

$f'(\pi/2) = 0$

$y - 1 = 0 \cdot (x - \pi/2)$

$\Rightarrow \boxed{y = 1}$

Product and Quotient Rules

4. Compute

$$\frac{\mathrm{d}}{\mathrm{d}x}\left[x\tan(x)\right].$$

$$\frac{d}{dx}\left[x\tan(x)\right] = \boxed{\tan(x) + x\sec^2(x)}.$$

5. Compute

$$\frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{\sin^2(x)}{\cos(x)}\right].$$

$$\frac{d}{dx}\left[\frac{\sin^2(x)}{\cos(x)}\right] = \frac{2\sin(x)\cos(x)\cos(x) - \sin^2(x)(-\sin(x))}{\cos^2(x)}$$

$$= \boxed{\frac{2\sin(x)\cos^2(x) + \sin^3(x)}{\cos^2(x)}}$$

Chain Rule

6. Compute

$$\frac{\mathrm{d}}{\mathrm{d}x}\left[\sin\left(x^3+7\right)\right].$$

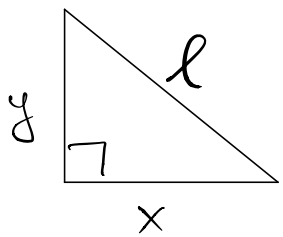
$$\frac{d}{dx}\sin(x^3+7) = \cos(x^3+7)\frac{d}{dx}(x^3+7)$$

$$= \cos(x^3+7)(3x^2)$$

$$= \boxed{3x^2\cos(x^3+7)}$$

Implicit Differentiation and Related Rates

7. The top of a ladder slides down a vertical wall at a rate of 3 meters/second. At the moment when the top of the ladder is 4 meters from the ground, it slides away from the wall at a rate of 4 meters/second. How long is the ladder?



$$\ell^2 = x^2 + y^2 \Rightarrow \ell = \sqrt{x^2+y^2}$$

Given: $dy/dt = -3\,m/s$

$\frac{dx}{dt} = 4\,m/s$ when $y = 4$

$\frac{d\ell}{dt} = 0$

$$\ell^2 = x^2 + y^2 \Rightarrow 0 = 2x\frac{dx}{dt} + 2y\frac{dy}{dt}$$

$$= 2x(4) + 2(4)(-3)$$

$$= 8x - 24$$

$$\Rightarrow 8x = 24$$

$$\Rightarrow x = 24/8 = 3$$

$$\Rightarrow \ell = \sqrt{3^2+4^2} = \sqrt{9+16}$$

$$= \sqrt{25} = \boxed{5m}$$

Derivatives and Shape

Use the function

$$f(x) = x^4 - 2x^3$$

to answer the following questions.

8. Find the intervals where the function is increasing and decreasing. Write NONE if not applicable.

Increasing: $(3/2, \infty)$

Decreasing: $(-\infty, 3/2)$

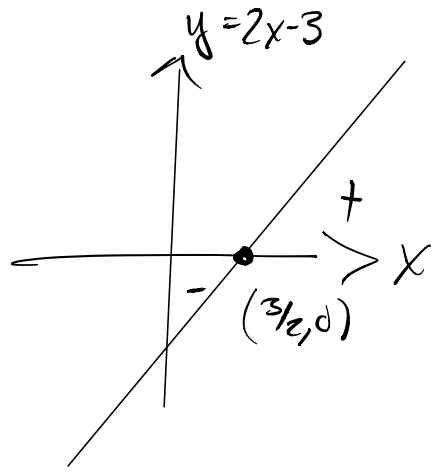
$0 = f'(x) = 4x^3 - 6x^2$

$= 2x^2(2x-3)$

$\iff x = 0$ or $2x - 3 = 0$

$\iff x = 0$ or $x = 3/2$

$2x^2 \geq 0$, so sign of $f'(x)$ depends on $2x-3$.



9. State the local maximum and local minimum value(s). Write NONE if not applicable.

Local maximum value(s): NONE

Local minimum value(s): $(3/2, -27/16)$

$f(x) = x^3(x-2)$

$f\left(\frac{3}{2}\right) = \left(\frac{3}{2}\right)^3\left(\frac{3}{2} - \frac{4}{2}\right) = \frac{27}{8}\left(-\frac{1}{2}\right) = -\frac{27}{16}$

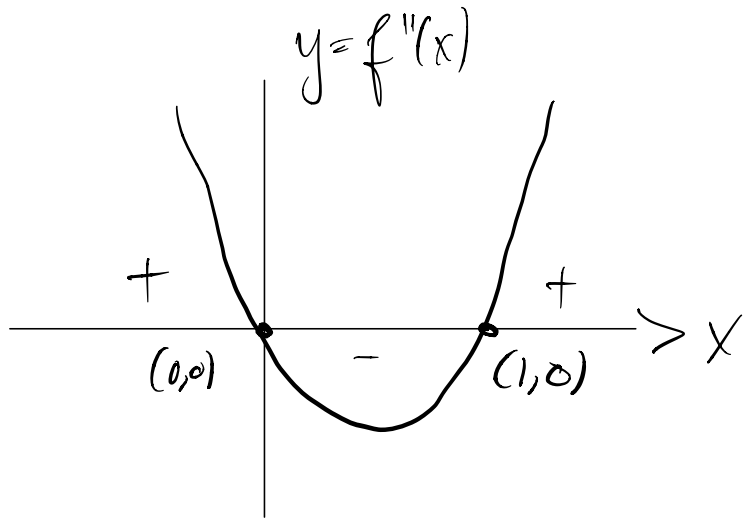
10. Find the intervals on which the function is concave up and concave down. State the inflection points. Write NONE if not applicable.

Concave Up: $(-\infty, 0) \cup (1, \infty)$

Concave Down: $(0, 1)$

Inflection Points: $(0,0)$, $(1,-1)$

$$f''(x) = 12x^2 - 12x = 12x(x-1)$$



$$f(1) = 1^4 - 2(1)^3 = 1 - 2 = -1.$$

Asymptotes

11. Find the asymptotes of

$$f(x) = \frac{3x^2}{x^2 - 4}$$

Write NONE if there are none.

Horizontal: $y=3$

Vertical: $x=2,\ x=-2$

$$\lim_{x\to\infty} \frac{3x^2}{x^2-4} = \lim_{x\to\infty} \frac{x^2}{x^2}\left(\frac{3}{1-4/x^2}\right)$$

$$= \lim_{x\to\infty} \frac{3}{1-4/x^2}$$

$$= \frac{3}{1-0} = 3$$

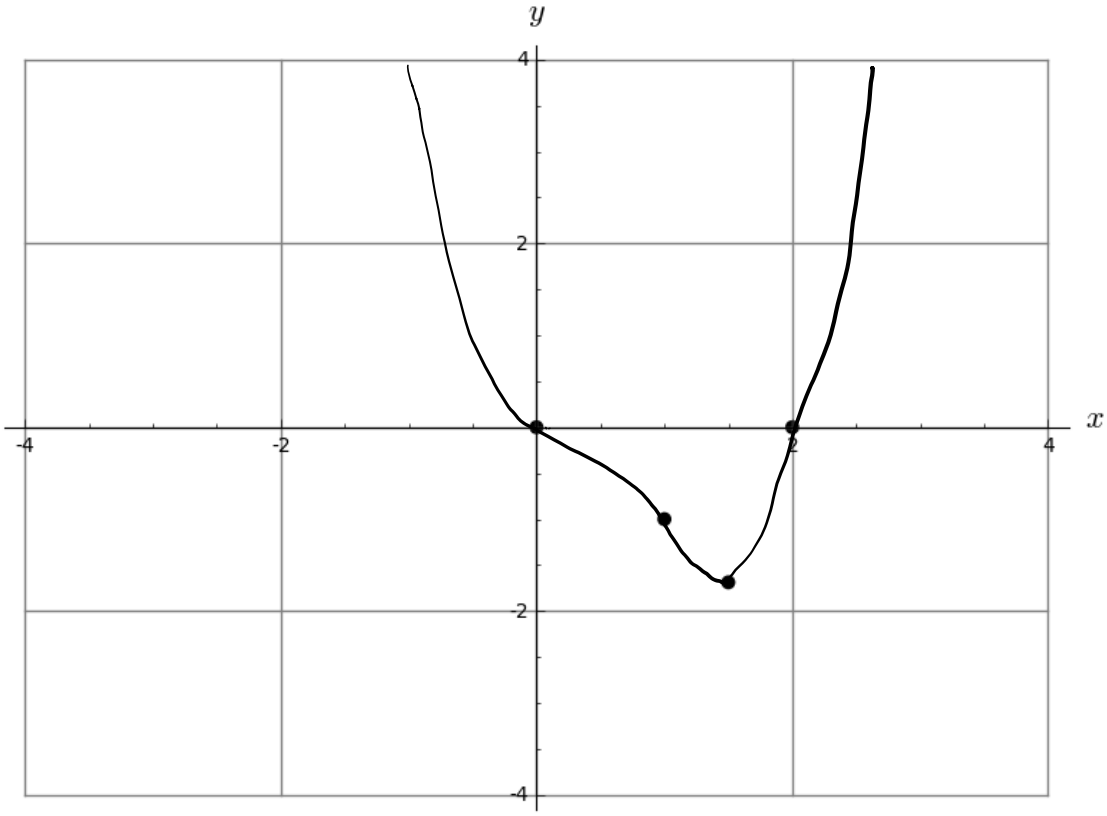
$$\lim_{x\to-\infty} \frac{3x^2}{x^2-4} = \lim_{x\to\infty} \frac{3}{1-\frac{4}{x^2}} = \frac{3}{1-0} = 3$$

Curve Sketching

12. You are given the following information about a function, f:

- x-intercepts: $(0,0)$ and $(2,0)$,
- y-intercepts: $(0,0)$,
- asymptotes: none,
- critical points: $(0,0)$ and $\left(\frac{3}{2}, -\frac{27}{16}\right)$,
- increasing: $\left(\frac{3}{2}, \infty\right)$,
- decreasing: $(-\infty, 3/2)$,
- concave up: $(-\infty, 0) \cup (1, \infty)$,
- concave down: $(0,1)$, and
- $f(1) = -1$.

Use this information to sketch the curve.



Closed Interval Method and Optimization

13. Find the absolute maximum and minimum values of $f(x) = x^3 - 6x^2 + 9x$ on $[-1, 4]$.

$$f(x) = x^3 - 6x^2 + 9x$$
$$= x(x^2 - 6x + 9)$$
$$= x(x-3)^2$$
$$f'(x) = 3x^2 - 12x + 9$$
$$= 3(x^2 - 4x + 3)$$
$$= (x-1)(x-3)$$
$$= 0$$
$$\Leftrightarrow x = 1,\ x = 3$$

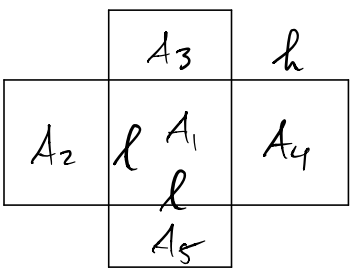
$f(-1) = -(-1-3)^2 = -(-4)^2 = \boxed{-16}$ min

$f(1) = 1(1-3)^2 = (-2)^2 = \boxed{4}$ max

$f(3) = 3(3-3)^2 = 0$

$f(4) = 4(4-3)^2 = \boxed{4}$ max

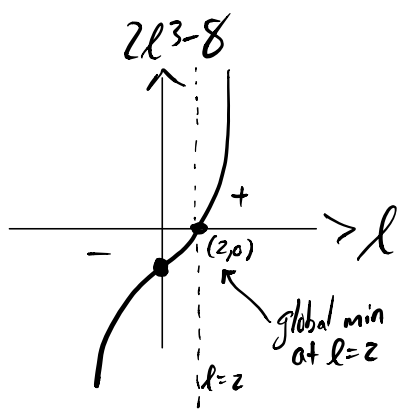
14. A box with a square base and no top must have a volume of $4\,\text{cm}^3$. Find the dimensions of the box that minimize the amount of material used.



Surface area $= S = A_1 + A_2 + A_3 + A_4 + A_5$

$$A_1 = \ell^2$$
$$A_2 = A_3 = A_4 = A_5 = \ell h$$
$$\Rightarrow S = \ell^2 + 4\ell h$$
$$= \ell^2 + 4\ell(4/\ell^2)$$
$$= \ell^2 + 16/\ell,\ \ell \neq 0$$

$V = 4 = \ell^2 h$

$\Rightarrow h = 4/\ell^2,$

$$0 = S' = 2\ell - 16/\ell^2 = \frac{2\ell^3 - 16}{\ell^2}$$
$$\Leftrightarrow 2\ell^3 - 16 = 0$$
$$\Leftrightarrow \ell^3 = \frac{16}{2} = 8$$
$$\Leftrightarrow \boxed{\ell = 2,\ h = 4/2^2 = 1}$$